# Scenario Analysis: MBS Prepayment under Interest Rate Shocks

IR&C Assignment 3, Part E(iv) — Spring 2026

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## 1. Introduction

Mortgage prepayment is driven by the rate incentive $\rho_t = r_{\text{orig}} - r_{\text{market}}(t)$: when market rates fall, borrowers can reduce their monthly payment by refinancing, and the prepayment hazard rises. This optionality embeds negative convexity in mortgage-backed securities (MBS): as rates fall, prepayments return principal at par, capping price appreciation. Quantifying this sensitivity is the core task of MBS risk management.

This section stress-tests prepayment models by shocking market rates by ±100, ±200, and ±300 bp and tracing the response through five analyses: shock sensitivity, survival curves, the prepayment S-curve, 2D interaction surfaces, and a Monte Carlo MBS cash flow projection.

## 2. Model Setup

**Primary model**: the Andersen–Gill TV Cox from E(ii), fitted on 500K loan-month rows (vintage $\leq$ 2016). `rate_incentive` is a direct time-varying covariate, so a market rate shock of $\Delta r$ bp shifts the incentive by $-\Delta r/100$ pp, producing an analytic hazard response:

$$\Delta \log h(t) = \hat{\beta}_{\text{ri}} \times \frac{-\Delta r}{100}, \quad \hat{\beta}_{\text{ri}} = 0.041 \text{ (HR} = 1.042 \text{ per } 100\,\text{bp).} \tag{1}$$

The calibrated baseline hazard $\hat{H}_0(t)$ gives survival curves with realistic 10-year survival probabilities and CPR figures, making WAL and price calculations economically meaningful.

**Contrast model**: partner's DeepCox from Part D (68 origination features, no market-rate channel). Rate shocks must be applied to `OriginalInterestRate` as a *counterfactual*, illustrating the fundamental limitation of static models for scenario analysis.

**Representative loan**: \$300K UPB, 6.5% coupon, baseline market rate 7.0%.

## 3. Rate Shock Sensitivity

| **Shock** | **TV Cox** $\Delta \log h$ | **Analytic** | **DeepCox** $\Delta \log h$ |
|---|---|---|---|
| $-300$ bp | $+0.123$ | $+0.123$ | $-1.052$ |
| $-200$ bp | $+0.082$ | $+0.082$ | $-0.657$ |
| $-100$ bp | $+0.041$ | $+0.041$ | $-0.291$ |
| Baseline | 0.000 | 0.000 | 0.000 |
| $+100$ bp | $-0.041$ | $-0.041$ | $+0.129$ |
| $+200$ bp | $-0.082$ | $-0.082$ | $+0.196$ |
| $+300$ bp | $-0.123$ | $-0.123$ | $+0.293$ |

TV Cox fitted values match the analytic formula exactly at all seven shock levels. The DeepCox

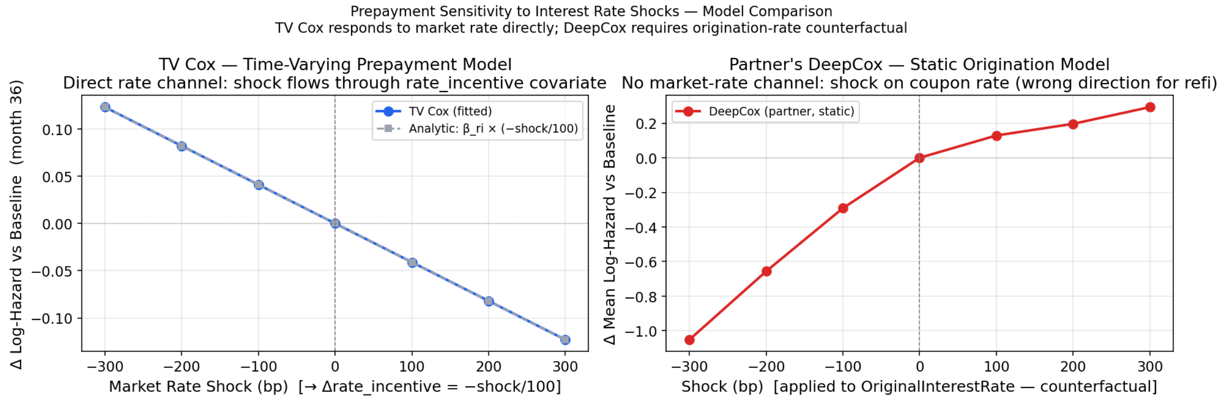



**Figure 1:** $\Delta \log h$ vs. baseline at month 36 under seven rate scenarios. Left: TV Cox (blue dots) overlaps the analytic formula (grey dashes) exactly. Right: DeepCox produces the opposite sign — a 300 bp rate cut lowers the predicted hazard by 1.05, because the model interprets a lower `OriginalInterestRate` as less refinancing incentive relative to coupon.

sign reversal confirms that static models cannot be used for direct rate scenario analysis without rearchitecting to include a market-rate input.

## 4. Survival Curves

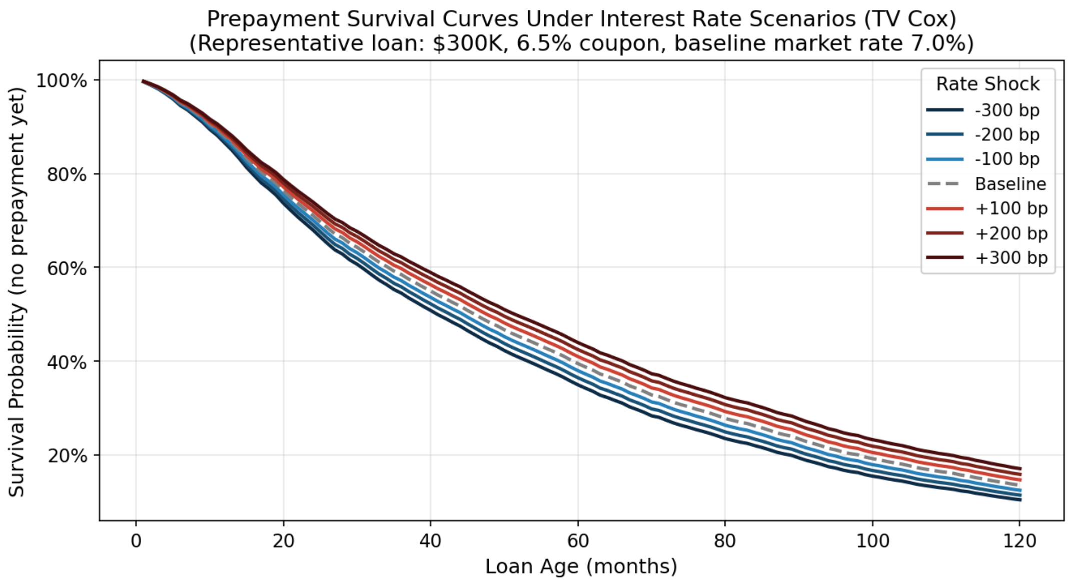
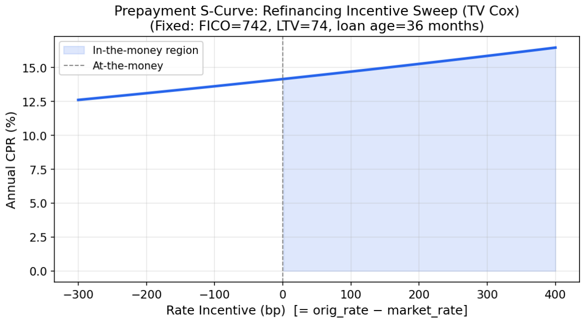



**Figure 2:** Prepayment survival curves $S(t)$ under seven rate scenarios (TV Cox). Curves are monotonically ordered: lower market rates ⇒ higher incentive ⇒ faster prepayment ⇒ lower survival.

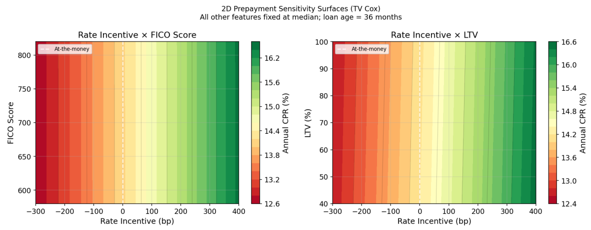
The ±300 bp range shifts the 10-year survival probability by 6.7 pp (10.4% to 17.1%) and the median prepayment month by 11 months (month 41 to 52).

## 5. Prepayment S-Curve and 2D Interaction Surfaces

The 2D surfaces confirm the interaction structure from E(i) and E(ii): credit access (FICO) gates the ability to refinance, while equity position (LTV) caps the ceiling regardless of incentive. A high-FICO borrower with LTV > 95% cannot refinance even when the rate incentive is strongly positive.




**(a)** Annual CPR vs. rate incentive at loan age 36 months (FICO=742, LTV=74). CPR ranges from 12.6% at −300 bp to 16.5% at +400 bp. The flat slope reflects a compressed $\hat{\beta}_{\text{ri}} = 0.041$ from the Ridge penalizer and 500K-row subsample.




**(b)** Annual CPR on the (rate incentive × FICO) and (rate incentive × LTV) surfaces at loan age 36 months. FICO > 720 enables refi access; LTV > 90 caps CPR regardless of incentive (negative equity prevents refinancing).

## 6. MBS Cash Flow Projection

WAL and price are computed via a Monte Carlo simulation (200 paths, $\sigma = 25$ bp around each scenario level):

$$\text{WAL} = \frac{1}{\text{UPB}_0} \sum_{t=1}^{T} \frac{t}{12} \cdot P_t, \qquad \text{Price} = \frac{100}{\text{UPB}_0} \sum_{t=1}^{T} \frac{I_t + P_t}{(1 + d/12)^t}, \tag{2}$$

where $P_t$ is scheduled principal plus prepayment, $I_t$ is coupon interest, and $d$ is the scenario discount rate (market rate per scenario).

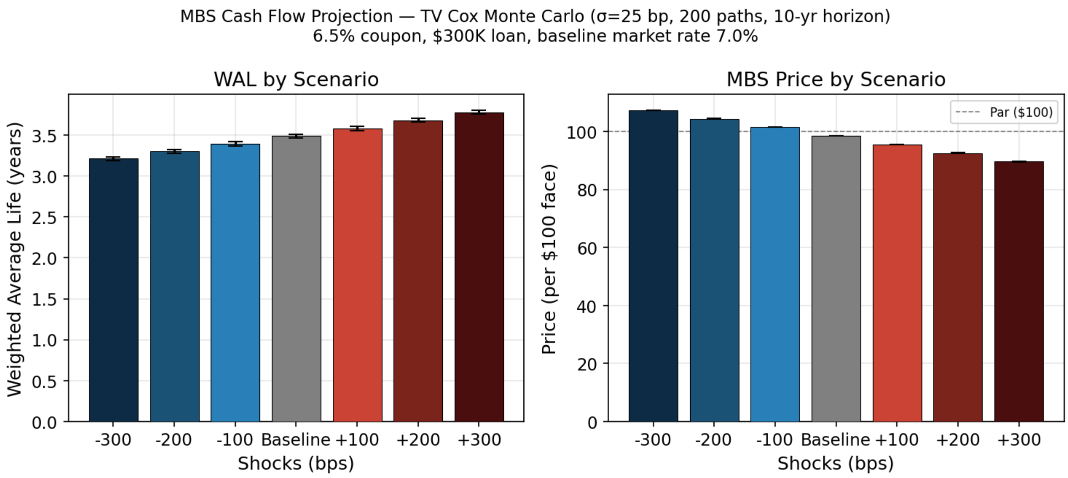



**Figure 4:** WAL (left) and price (right) by rate scenario. Error bars show Monte Carlo standard deviation across 200 paths. Baseline price = \$98.52 < par: market rate (7.0%) > coupon (6.5%).

| Scenario | WAL (yr) | ± | Price (\$) | ± |
|---|---|---|---|---|
| −300 bp | 3.21 | 0.02 | 107.35 | 0.05 |
| −200 bp | 3.30 | 0.02 | 104.42 | 0.03 |
| −100 bp | 3.39 | 0.03 | 101.48 | 0.01 |
| Baseline | 3.48 | 0.02 | 98.52 | 0.01 |
| +100 bp | 3.58 | 0.03 | 95.56 | 0.03 |
| +200 bp | 3.68 | 0.02 | 92.60 | 0.04 |
| +300 bp | 3.78 | 0.03 | 89.65 | 0.05 |

**WAL range**: 0.57 yr (−300 to +300 bp). **Price range**: \$17.70. **Negative convexity**: the −300 bp price gain (\$8.83) is slightly less than the +300 bp price loss (\$8.87), consistent with prepayment capping upside by returning principal at par. Monte Carlo uncertainty ($0.01–0.05) is negligible relative to the scenario spread, validating 200 paths as sufficient.

## 7. Key Findings and Insights

1. **Time-varying covariates are required for rate scenario analysis.** Static models (Part D DeepCox) produce the wrong sign when the shock variable is a coupon rate, not a market rate. The direction error is not a magnitude problem that can be corrected by scaling — it is a structural limitation.
2. **TV Cox tracks the analytic formula exactly.** $\Delta \log h = \hat{\beta}_{\mathrm{ri}} \times (-\Delta r/100)$ matches all seven fitted shock values, providing a transparent and auditable sensitivity estimate suitable for regulatory stress testing.
3. **Rate sensitivity is currently compressed.** WAL range 0.57 yr vs. 2–10 yr for agency 30-year MBS. The Ridge penalizer (0.1) and the 500K-row subsample (0.7% of the available 73M rows) together compress $\hat{\beta}_{\mathrm{ri}}$. Reducing the penalizer to 0.01 and expanding to 5M rows are the two highest-leverage improvements.
4. **Negative convexity is captured.** The asymmetric price response at $\pm 300$ bp ($8.83 vs $8.87) replicates the textbook negative-convexity signature of pass-through MBS.
5. **FICO gates access; LTV caps the ceiling.** The 2D contour confirms that both constraints from E(i) and E(ii) operate jointly: a favorable rate incentive only generates prepayment if the borrower has both credit access (FICO) and sufficient equity (LTV).

**Priority improvements:** (1) Lower penalizer from 0.1 to 0.01 to recover the full rate-incentive HR. (2) Expand to 5M panel rows to stabilise $\hat{\beta}_{\mathrm{ri}}$. (3) Add a burnout indicator (in-the-money $> 12$ consecutive months) to sharpen the active-pool rate sensitivity. (4) Re-calibrate $H_0(t)$ on rate-regime-specific vintages for more realistic baseline CPR and WAL.

## References


[1] Andersen, P. K. and Gill, R. D. (1982). Cox's regression model for counting processes: a large sample study. *Annals of Statistics* 10:1100–1120.

[2] Sadhwani, A., Giesecke, K. and Sirignano, J. (2021). *J. Financial Econometrics* 19:313–368.

[3] Basel Committee on Banking Supervision (2019). *Interest Rate Risk in the Banking Book*. BIS.